

PCT-Like Syndromes and Differential Considerations

PCT-like syndromes have been reported in association with hepatic tumors and systemic lupus erythematosus. Subepidermal bullous dermatoses that clinically and histologically mimic porphyria cutanea tarda (PCT) have also been described (see Pseudoporphyria). True PCT has been observed in patients with renal failure undergoing hemodialysis.

Laboratory Findings in PCT

Confirmation of PCT relies on detecting markedly elevated plasma porphyrin levels—typically 5- to 100-fold higher than normal—and increased fecal excretion of isocoproporphyrin (ISOCOPRO).

Urinary Porphyrin Excretion

Patients with PCT excrete increased amounts of porphyrins in the urine. In some cases, urine may exhibit a pink-red fluorescence under a Wood's lamp, although this finding is not always present.

The urinary porphyrin profile in PCT is characterized by three main features:

1. Elevated excretion of uroporphyrin (URO) and other acetate-substituted porphyrins.
2. A distinct isomer pattern of porphyrins, with mixtures of type I and type III isomers.
3. Increased fecal excretion of ISOCOPRO.

Urinary porphyrins in PCT include:

- Markedly increased 8-carboxyl URO
- Elevated 7-, 6-, and 5-carboxyl porphyrins
- Moderately increased 4-carboxyl porphyrin (COPRO), typically not exceeding 600 µg per 24 hours

The 8- and 7-carboxyl porphyrins constitute more than 90% of total urinary porphyrins. The isomer distribution is as follows:

- URO: ~60% type I, ~40% type III
- 7- and 6-carboxyl porphyrins: >90% type III, <10% type I
- 5- and 4-carboxyl porphyrins: ~50% each of type I and type III

This specific isomer pattern is consistently observed in PCT and aids in diagnosis.

Fecal Porphyrin Excretion

In healthy individuals, uroporphyrin is present in stool only in trace amounts. In PCT, fecal porphyrin levels are elevated and primarily consist of: - Isocoproporphyrin (ISOCOPRO), predominantly type III - 7-carboxyl porphyrin - Smaller amounts of URO and COPRO

Total 24-hour fecal porphyrin excretion may exceed urinary excretion in PCT patients.

Differentiation from Variegate Porphyria (VP)

The urinary URO-to-COPRO ratio is a useful diagnostic tool: - In PCT: ratio is typically $>3:1$ - In VP: ratio is usually $<1:1$

In rare cases, patients with classic cutaneous features of PCT may have normal or only mildly elevated 24-hour urine porphyrin levels. In such instances, stool porphyrin analysis is essential, as fecal porphyrins are elevated in VP.

Diagnostic Pitfalls and Confirmatory Testing

A negative urine fluorescent screening test does not rule out PCT. Some patients with clinical PCT may have normal screening results. Therefore, definitive diagnosis requires: - Measurement of plasma porphyrin levels - Quantitative 24-hour urine URO and COPRO analysis - Stool PROTO and COPRO determinations

High-performance liquid chromatography (HPLC) is preferred for accurate porphyrin speciation and differentiation between PCT and VP.

In rare cases of VP with normal fecal porphyrin excretion, bile porphyrin analysis may provide additional diagnostic clarity.

Iron Overload in PCT

Virtually all patients with PCT exhibit iron overload, evidenced by: - Elevated serum iron - Increased ferritin levels - Hepatocellular iron accumulation

Mild erythrocytosis may occasionally be present.